Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider a sequence of symbols emitted by a source with memory

$$\{2, 1, 1, 3, 5, 2, 1, 0\} \tag{1}$$

   (a) Determine the Lloyd-Max quantizer with $R = 1$ bit / symbol. The initial intervals are $I_1 = (-\infty, 2]$ and $I_2 = (2, +\infty]$.

   (b) Encode the sequence showing the decoded symbols and the corresponding MSE.

   (c) Compute the empirical entropy of the quantization symbols.

   (d) Compute the prediction residuals in open loop. Based on this result, compute the Lloyd-Max quantizer. The initial intervals are $I_1 = (-\infty, 0]$ and $I_2 = (0, +\infty]$.

   (e) Encode the sequence with DPCM showing the decoded symbols and the corresponding MSE (assume that the first symbol is lossless coded).

2. **Question (5 pts)**
   (a) Illustrate Elias coding for the case of a binary memoryless source.

   (b) Describe how Elias coding can be adapted to take into account time-varying statistics.

3. **Question (5 pts)**
   (a) Illustrate the block diagram of an image coding system. Briefly describe the input and output of each block.

   (b) Comment on the computational complexity of the different blocks.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider a transform coding system, where T is the $M \times N$ transform matrix.
   - [T] [F] When $M = N$, the transform is critically sampled.
   - [T] [F] Let $M = N$. $\sum_{i=1}^{N} \sigma_{x_i} = \sum_{i=1}^{N} \sigma_{y_i}$
   - [T] [F] Let R_i denote the rate allocated to the ith coefficient. Under optimal bit allocation, $R_i = R + c \log_2 \sigma_{y_i}^2$
   - [T] [F] The KLT is the transform that minimizes $\frac{1}{N} \sum_{i=1}^{N} \sigma_{y_i}$.

   (b) Consider an audio coding system
   - [T] [F] For some critical bands, it might happen that SMR is negative.
   - [T] [F] Bit allocation is inversely proportional to the SMR
   - [T] [F] The MDCT achieves perfect reconstruction.

   (c) Let X denote a discrete source with memory.
   - [T] [F] It is possible to design a code whose average length $L < H(X)$
   - [T] [F] $H(x_1, \ldots, x_N)/N \geq H(x_N | x_{N-1}, \ldots, x_1)$
   - [T] [F] $L_H \leq L_S$, where L_S and L_H denote, respectively, the average length of Shannon and Huffman coding.

5. **Question (Matlab - 10pts)**

   Load the dataset `data.m`.
   (a) Let $\{z_i\}_{i=1}^{M}$ denote the sequence of symbols to be encoded. Let $\{x_i\}_{i=1}^{M/2}$ and $\{y_i\}_{i=1}^{M/2}$ denote, respectively, the sequence of symbols at odd and even time instants. Plot the samples in the $x - y$ plane.

   (b) Perform scalar quantization with $R = 1$ bits / symbol. Design the optimal quantizer for both $\{x_i\}_{i=1}^{M/2}$ and $\{y_i\}_{i=1}^{M/2}$.

   (c) Perform optimal vector quantization with $R = 1$ bits / symbols, coding 2-dimensional vectors $[x_i, y_i]^T$, $i = 1, \ldots, M/2$.

   (d) Compare the SNR in the two cases.

   (e) Plot the reconstruction levels in both cases, superimposed to the data samples on a 2-dimensional plane.